Response to Reviewers

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Duplicate-Aware Validation of Machine-Learning Models for Recorded Sleep-Status Labels in a Public Lifestyle Dataset

We thank the editor and reviewers. The revision changes both the analysis and the manuscript. We verified that "None" is a literal source-file category, identified extensive duplicate predictor profiles, replaced the single random hold-out analysis with repeated duplicate-aware validation, released executable code and observation-level predictions, removed unsupported explainability figures, and narrowed the work to a reproducibility-focused benchmark audit. The earlier 97.33% single-split result is not retained as a primary finding.

Reviewer 1

1. **Outcome coding.** The CSV is now read with missing-value conversion disabled and the code stops if an empty or undocumented outcome occurs. The audit finds 219 literal "None", 77 "Insomnia", and 78 "Sleep Apnea" labels, with no blank outcomes. We do not interpret "None" as a clinically adjudicated absence of disease (Methods, Data audit; Table 1).

2. **Primary configuration.** The revised primary analysis uses only RF+XGBoost stacking with a logistic meta-learner. SVM is removed. Soft voting uses exactly the same two base learners.

3. **Reproducibility.** `revised_analysis.py` is the executable analysis. It exports software versions, split/group assignments, fold metrics, reference labels, predictions, and probabilities. Preprocessing is refit inside every inner stacking fold.

4. **Validation and comparisons.** Two shuffled repetitions of five-fold StratifiedGroupKFold replace the single hold-out. Exact retained-predictor duplicates cannot cross a fold. Stacking, soft voting, both base learners, logistic regression, a shallow tree, and histogram gradient boosting use identical outer folds.

5. **Statistics.** The released predictions permit reproduction of all comparisons. Stacking minus soft-voting mean accuracy is -0.0031 (bootstrap 95% interval -0.0162 to 0.0070); the first-repeat out-of-fold McNemar result is $p = 0.6875$. No superiority claim remains.

6. **Explanations.** SHAP, LIME, PCA, and ALE results were removed rather than presenting unstable attributions from a duplicate-rich, conflicting-label dataset.

Reviewer 2

1. We no longer claim novelty for stacking. The contribution is the audit and duplicate-aware reproducible evaluation protocol.

2. Risk-stratification, clinical-decision-support, and diagnostic claims were removed. The manuscript describes prediction of recorded source-file labels only.

3. The single 75-observation result is replaced by repeated grouped validation, variability, probability scores, calibration artefacts, and paired comparisons.

4. Post-hoc explanations are removed and are not used as biological mechanisms.

5. Histogram gradient boosting was added as a contemporary tabular comparator.

6. The paper was rewritten, condensed, and regenerated from the revised results. The suggested traffic forecasting, image segmentation, and remote-sensing papers are not topical to sleep-health tabular modelling or this methodology, so they were not cited.

Reviewer 3

1. The Discussion explicitly separates internal benchmark discrimination from requirements for real clinical screening.

2. Results explain the small number of discordant predictions, the interval crossing zero, and limited paired-test power.

3. Rather than speculate about correlated-feature attribution rankings, the attribution results were removed.

4. The inconsistent RF+XGBoost+SVM robustness analysis is removed.

5. Limitations now state the potential classification bias and unknown clinical meaning of the literal "None" category.

6. Local-explanation figures are removed; released observation-level predictions retain the information needed for auditability.

7. The paper recommends prospective cohorts, development-only tuning, calibration assessment, and external validation.

8. The suggested references were reviewed for relevance and were not added because they are outside the paper's topic.

Reviewer 4

1. We verified source outcomes and report 265 duplicate retained-predictor rows, 109 unique profiles, and 22 profiles with conflicting recorded outcomes. Group validation is now mandatory.

2. The supplied code generates every reported result and releases package versions, groups, splits, probabilities, metrics, and paired statistics.

3. Person ID is removed before group construction. Specifications are fixed; no test-fold tuning is performed.

4. Repeated grouped validation, simple baselines, Brier scores, and out-of-fold reliability curves were added.

5. SHAP/LIME/ALE material was removed from both paper and title.

Reviewer 5

1. The raw import, literal-label check, cleaning, and outcome counts are implemented in `load_data()` and documented in Table 1.
2. The 97.33% 75-observation score is no longer used as evidence; ten duplicate-aware outer folds and their variation are reported.
3. Code, versions, seeds, group splits, metrics, and observation-level predictions are included in the repository.
4. The RF+XGBoost two-learner configuration is used consistently throughout; SVM has been removed.

Files supplied

- `revised_submission.tex`, `revised_references.bib`, and compiled manuscript PDF;
- this response and the editable `response_to_reviewers.md`;
- `revised_analysis.py`, `revised_stacking_cls.ipynb`, and `requirements.txt`;
- `revision_results/` containing metadata, exact splits, fold metrics, predictions/probabilities, paired statistics, and calibration artefacts.